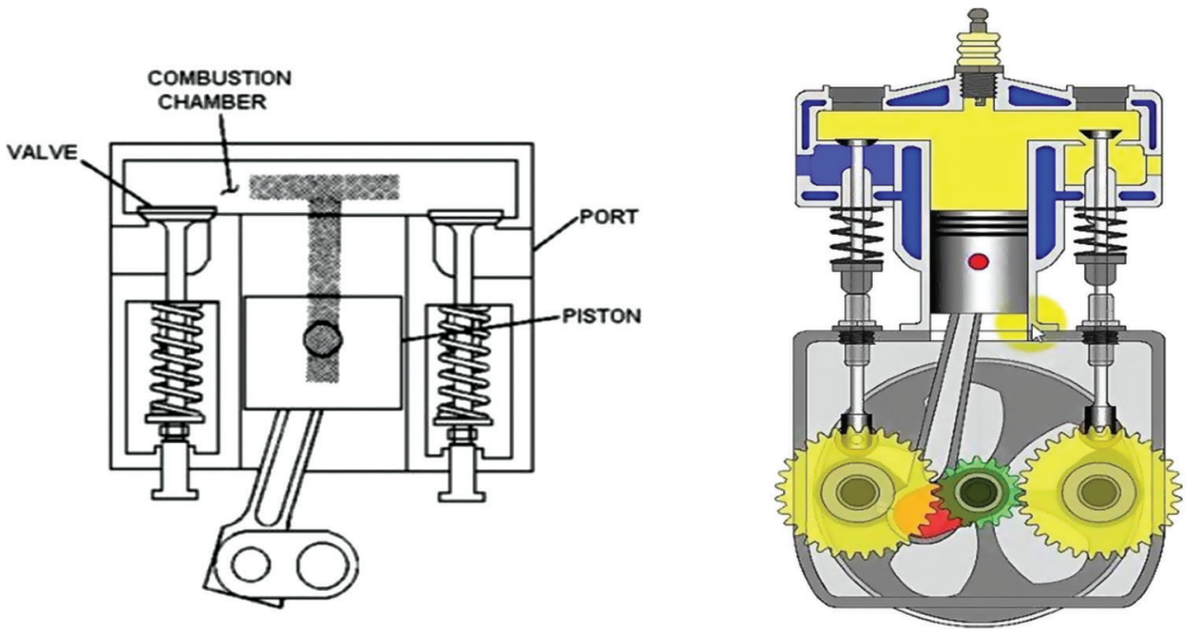



**Figure 6.11(d): T – Head Engine**

head in which the valves move in upward direction and the spark plug is fitted at the top of cylinder head. Figure 6.11(d) shows the 'T' Head Engine.

### Spherical shape

In engine's combustion chamber, if the cylinder head or the piston top has internally or externally a spherical shape then the combustion chamber can be called as spherical combustion chamber. The inlet and exhaust valves are fitted at the opposite direction and the spark plug is fitted either centre or at lateral sides in such engines.

### 6.11.1 Types of combustion chambers of diesel engines

Due to high power output of diesel engines, the combustion chambers are designed in different types depending on the engines torque and speed requirements. The diesel engine combustion chambers are mainly of the following,

a. Open Combustion chamber
b. Pre-Combustion chamber
c. Swirl Combustion chamber
d. Squish Combustion chamber
e. Air cell and energy cell

### (a) Open Combustion chamber

This type of combustion is mainly used in slow and high-speed engines. The piston head is designed in such a way to have a (semi spherical) bowl shape and the injector of the engine is mounted on the cylinder head. Figure 6.11.1(a) shows the Open Combustion Chamber.

### (b) Pre-Combustion chamber

It is mainly used in high speed engines. It consists of two chambers such as auxiliary (or pre) combustion chamber and other called as the main combustion chamber. The auxiliary combustion chamber is small and is used for igniting the small amount of the air fuel mixture.



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